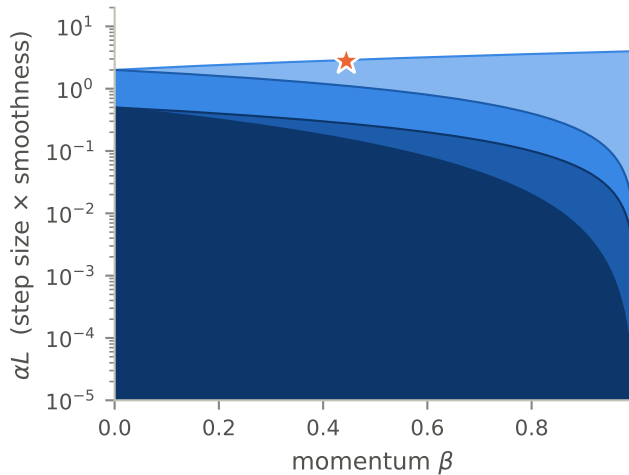
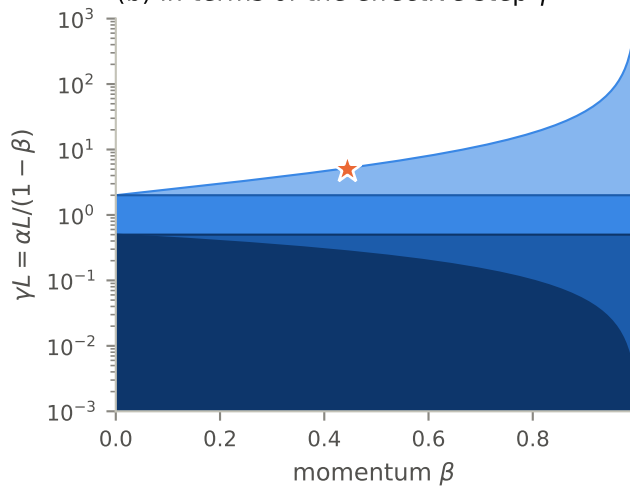


(a) in terms of the step size α



(b) in terms of the effective step γ



R4: quadratic f , exact gradients
R3: any smooth f , exact gradients

R2: convex f , stochastic
R1: nonconvex f , stochastic

★ LRP counterexample